Problem CF1. Suppose that we want to place n distinguishable books in a bookcase. We don't care how many shelves there are; we just want to make sure that each shelf has at least one book. The shelves are distinguishable and the order of books matters. Let $h(n)$ be the number of ways we can do this. Find $E_h(x)$, extract a formula for $h(n)$, and prove it combinatorially.

Problem CF2. Redo Problem CF1, now supposing the shelves are indistinguishable.

Problem CF3. Redo Problem CF1, now supposing the shelves are distinguishable but that the order of books on each shelf doesn't matter.

Problem CF4. Redo Problem CF1, supposing the shelves are indistinguishable and that the order of books on each shelf doesn't matter.

Problem CF5. Let $h(n)$ be the number of permutations in which every cycle has odd length. Show that

$$E_h(x) = \left(\frac{1+x}{1-x}\right)^{1/2}.$$

Problem CF6. Recall that the (unsigned) Stirling number of the first kind $s(n,k)$ is the number of permutations in $\mathfrak{S}_n$ with k cycles. Prove the lovely formula

$$\sum_{n\geq 0}\sum_{k=1}^{n} s(n,k)q^k\frac{x^n}{n!} = (1-x)^{-q}.$$

Problem CF7 (Stanley EC2 Problem 5.11(a)). Let $r(n)$ be the number of permutations $w \in \mathfrak{S}_n$ that are perfect squares, i.e., such that $w = x^2$ for some (not necessarily unique) $x \in \mathfrak{S}_n$. Determine $E_r(x)$.